
Craftsy

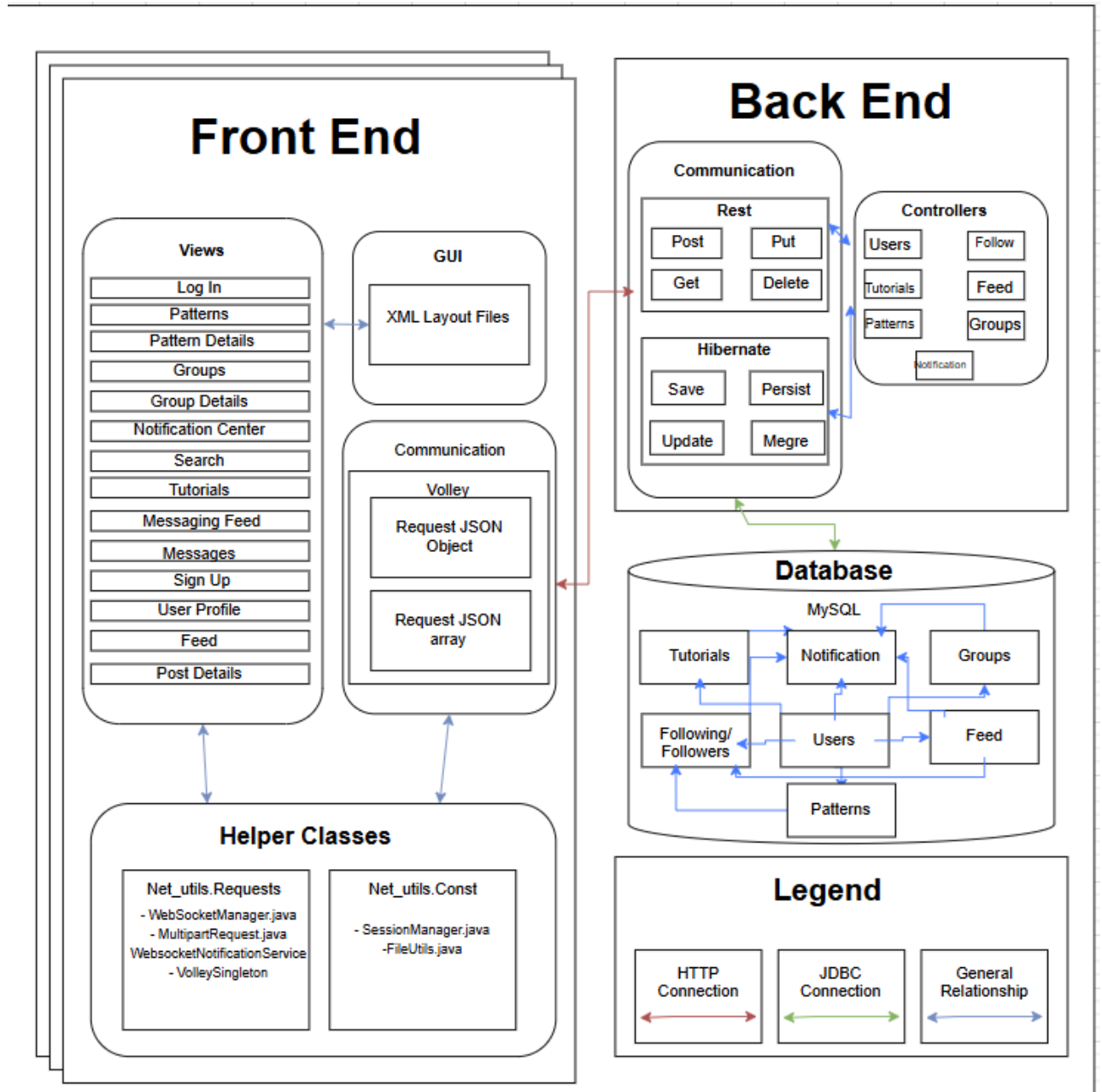
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Frontend

The frontend contains multiple views that represent the main interactions of Craftsby, including login, sign-up, feed, search, tutorials, patterns, groups, messaging, user profiles, and notifications. Each view communicates with the backend through structured JSON requests that follow a uniform request format.

To support dynamic content across these screens, we use Item classes to represent data objects displayed in lists, such as feed posts, tutorials, messages, group members, or pattern previews. These items provide a clean model that the UI can render.

Adapter classes serve as the bridge between these Item objects and the RecyclerViews that display them. They handle inflating layouts, binding data to screen elements, managing click listeners, and updating lists when backend data changes. This design keeps the views lightweight, ensures consistent UI behavior across all list-based features, and cleanly separates data representation from display logic.

Helper classes manage session persistence, file handling for tutorial uploads, and unified request construction. These abstractions ensure that actions such as posting to the feed, rating patterns, or joining groups all follow standardized communication patterns.

Backend

SERVER: WE UTILIZE SPRING BOOT WITH APACHE TOMCAT AS OUR SERVER. OUR SERVER CONTAINS THREE PRIMARY COMPONENTS:

- **Controllers** – Act as the middleman between the frontend and the backend. They handle incoming requests from the frontend, process them with business logic, and communicate with the database through repositories.
 - *Users*: The main entity that contains authentication details and basic user profile information. Users serve as the main entity for all interactions in the system. It consists of key attributes such as username, email, password, bio, craft specialties, and display name.
 - *Followers and Following*: Followers represent the relationship where a user can follow other users. Each follower record links a follower to a followed user. It supports one-to-many relationships where a user can have many followers. The following entity represents the reverse relationship, which allows a user to track who is following them.
 - *Notifications*: Stores messages intended to inform users about follower/Following status, social interactions, new tutorials posted, new tiers achieved, and other relevant updates.
 - *Tutorials*: Represent guide videos that are posted for other users. Each tutorial has attributes such as title and description, and it can only be posted by users who are in the Expert or Champion tier.
 - *Conversations*: Create group and direct conversations. This controller allows you to get all conversations a user is apart of. You can also add or remove users from group conversations. You can add or change a group convo picture.
 - *Patterns*: Can upload links to patterns and add information about them with photos. People can rate patterns and the total pattern rating will update. People can also leave comments on a pattern.
 - *Feed*: Users can upload projects they are working on. They can include photos and other information. Other users can comment on a users feed post. Any user can get their feed which has all projects posted by themselves or any of the users they follow.
- **Models (Entities)** – Represent the structure of our data and serve as a function storehouse with data fields, constructors, getters, and setters.
- **Repositories** – Serve as the communication layer between the server and the database. They handle CRUD operations for all entities and allow for complex queries, such as searching for challenges or filtering posts.

Database: Our database stores all information necessary for Craftsby to operate efficiently. We use MySQL, a relational database that supports a client-server architecture, allowing us to define relationships between data tables.

Database Schema

